UIQC Qiskit Fall Fest 2026: Final Project Report

Team Name: CAC (Collapse & Compute)

Team Members:

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Project Title: Sample, Then Diagonalize

1. Project Plan

1.1 Project Description:

Our project will explore sample-based quantum diagonalization (SQD) using the beginner notebook provided by the UIQC Club. Using a simplified quantum spin model, we aim to understand how measurement samples obtained from quantum circuits can be used to identify relevant bitstrings and estimate ground-state energy through classical diagonalization.

We will work through the notebook, examine the role of measurement sampling, and investigate how the number and selection of candidate bitstrings affects the resulting energy estimates. We will also discuss the limitations of the method and compare our observations with the reference results available in the notebook. As beginners in quantum computing, our goal is to develop a practical understanding of the relationship between quantum circuits, measurement, and classical computation.

Forked Repository: https://github.com/richardjoshua1942/UIQC-Qiskit-Fall-Fest-2026

1.2 Background and Objectives

As beginners in quantum computing, we aimed to bridge the gap between theoretical quantum mechanics and practical algorithm implementation. We selected the "Sample, Then Diagonalize" notebook to explore Sample-Based Quantum Diagonalization (SQD). SQD is a state-of-the-art hybrid quantum-classical algorithm that addresses the limitations of near-term quantum hardware. The primary objectives of this project were:

1. To understand the exponential scaling of quantum systems (the 2^nproblem) and the computational limits of classical diagonalization.
2. To investigate how quantum measurement probabilities concentrate on specific basis states.
3. To implement a classical subspace diagonalization function to extract ground-state energies from quantum samples.

4. To demonstrate that an unoptimized, shallow quantum circuit combined with classical post-processing can yield highly accurate energy estimates.

Division of Labor:

Both members collaborated on environment setup, Qiskit circuit execution, and data analysis. Richard focused on the classical numerical methods (np.linalg.eigvalsh), while Solomon focused on the quantum circuit parameter sweeps and probability distribution analysis.

2. Methodology

2.1 Classical Limitations and Exponential Scaling

The physical system modeled was a Transverse-Field Ising Chain, consisting of $n$qubits with nearest-neighbor interactions (ZZ) and a transverse magnetic field (X). This system is described by a Hamiltonian matrix (H). For $n$qubits, $H$is a $2^n \times 2^n$matrix. For $n = 8$, the matrix is 256×256. Extrapolating this scaling, storing the state vector of 50 qubits would require approximately 18 petabytes of classical memory. Therefore, finding the exact ground state via full classical diagonalization is computationally intractable for large systems.

2.2 Quantum Sampling

To bypass full diagonalization, a quantum circuit was utilized to prepare a trial state $|\psi\rangle$. The circuit was measured, and the probability of observing a specific bitstring $|b\rangle$is given by the squared amplitude: $P(b) = |\langle b | \psi \rangle|^2$. By executing the circuit for multiple shots, a set of the most probable bitstrings was collected.

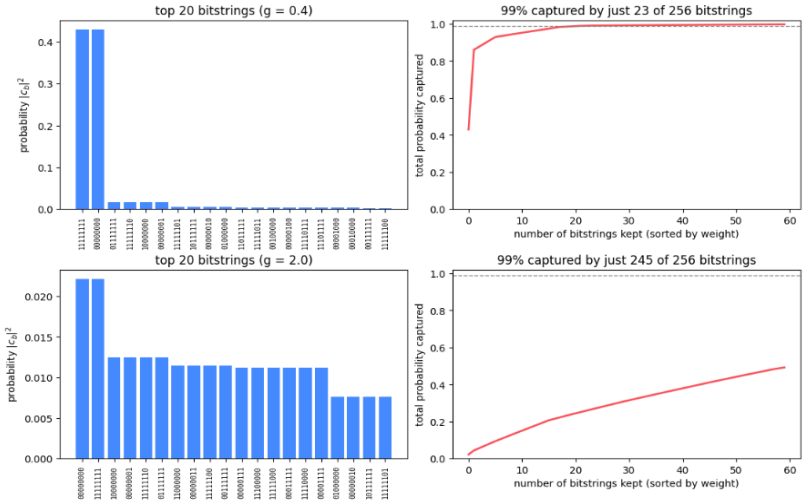
2.3 Subspace Diagonalization

Using the set of sampled bitstrings, the full Hmatrix was projected down to a subspace containing only the rows and columns corresponding to these states. The subspace_energy function was implemented to extract this sub-matrix using np.ix_ and calculate its lowest eigenvalue using np.linalg.eigvalsh. According to the variational principle, the lowest eigenvalue of this cropped matrix serves as an upper bound to the true ground-state energy.

3. Results

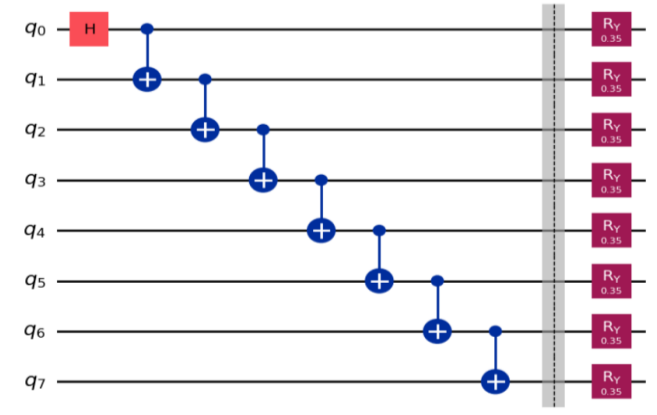
3.1 Probability Concentration

The probability distribution of the ground state was analysed at varying magnetic field strengths (g). At a low field ($g = 0.4$), the state was highly concentrated; we observed that 99% of the probability was captured by only 23 out of the 256 possible bitstrings. This validates that the relevant computational space can be drastically reduced.

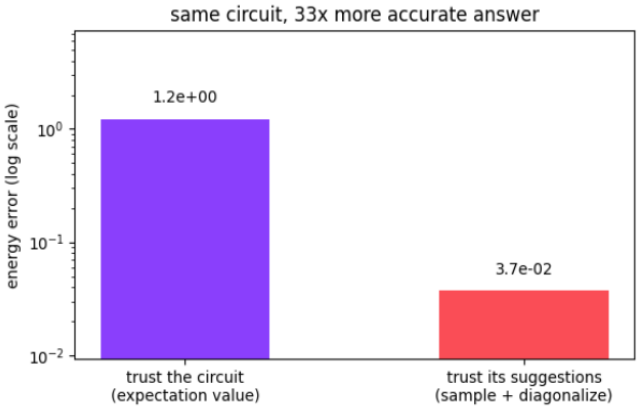


3.2 Performance of an Unoptimized Circuit

An unoptimized quantum circuit was constructed, consisting of a GHZ state (Hadamard and CNOT gates) with a rotational twist (ry, $\theta = 0.35$). This circuit lacked any variational optimization loop.

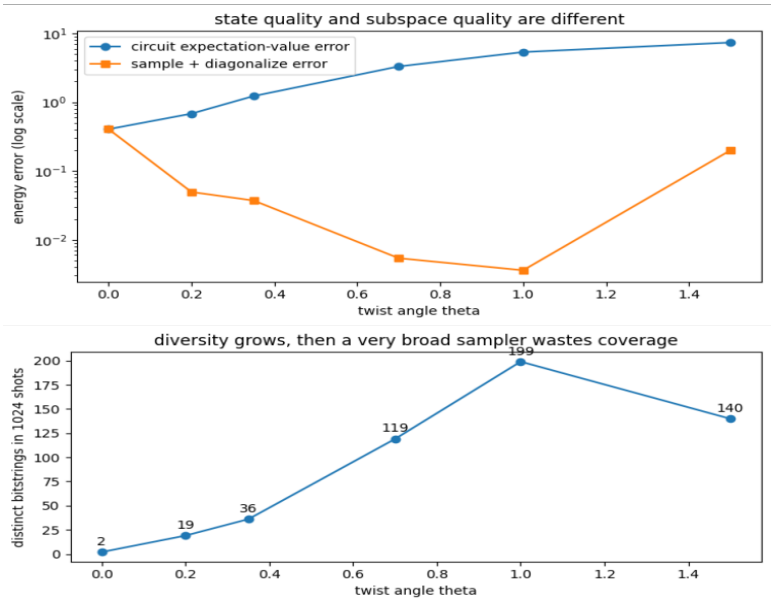


When the raw expectation value of this circuit was measured, the energy error was relatively high (1.23). However, when 1024 shots were sampled from the same circuit and processed through the classical diagonalization function, the energy error decreased significantly to 3.7×10^{-2}.



3.3 Parameter Sweep and Subspace Quality

We swept the rotation angle (θ) from 0.0 to 1.5 to observe the trade-off between circuit diversity and subspace quality.



At $\theta = 0.0$, the circuit produced only 2 distinct bitstrings, resulting in insufficient exploration of the Hilbert space. At $\theta = 1.5$, the circuit produced 140 distinct bitstrings; however, the probability was distributed too broadly, increasing the SQD error. The lowest error was observed at $\theta = 0.7$(119 distinct bitstrings), where the subspace provided sufficient support for the classical solver without introducing excessive noise.

4. Discussion

This project demonstrated the complementary roles within hybrid quantum-classical algorithms. While the unoptimized quantum circuit alone produced an inaccurate expectation value, it successfully generated a probability distribution concentrated on the relevant basis states. The classical computer then applied diagonalization to correct the projection coefficients within that restricted subspace, yielding an energy estimate close to the true ground state.

The parameter sweep revealed that the effectiveness of SQD is highly dependent on the sampling distribution. A circuit that is too uniform samples an insufficient number of states. Conversely, a circuit that is too random spreads the probability too thinly, which can overwhelm the classical solver with noise. The variational principle ensures that adding high-probability bitstrings to the subspace maintains or improves the energy estimate. Ultimately, this project confirmed that accurate ground-state energies can be approximated without full classical diagonalization by utilizing a quantum sampler.

5. Declaration of Assistance

In the spirit of AI-assisted learning, we declare the following resources and assistance used for this project:

- **AI Tools (ChatGPT and DeepSeek):** We used ChatGPT and DeepSeek as interactive tutors to explain the physical intuition behind Hamiltonians, the variational principle, and the mathematical difference between expectation values and sampled diagonalization. These tools were also used to debug environment-specific errors in Google Colab (such as `ModuleNotFoundError` and package installations)
- **Documentation:** The official Qiskit documentation and NumPy documentation for `np.ix_` and `np.linalg.eigvalsh`.
- **Provided Materials:** The `01_sample_then_diagonalize.ipynb` notebook and the `in-the-classroom with grader.zip` repository provided by UIQC.